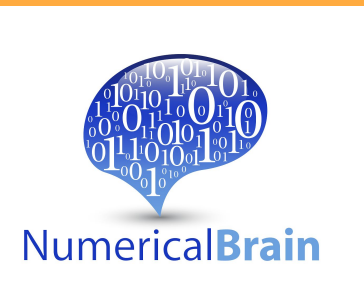


Real-Time Closed-Loop Brain-Body Simulation Using a GPU-Based Multi-Region Spiking Neural Network and a Mouse Musculoskeletal Model

GPUベース多領域 SNNとマウス筋骨格モデルを使ったリアルタイム閉ループ脳身体シミュレーション

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GPU-based SNN ran the brain-body loop faster than real time overall.

10.05 s simulated in 9.38 s; 50-ms blocks averaged 46.6 ms.

INTRO

- Motor control depends on closed-loop brain-body interactions with sensory feedback.
- We built a real-time closed-loop simulation by coupling a GPU-based multi-region SNN to a mouse forearm model

METHODS

- A GPU SNN simulated M1/S1, BG, TH, cerebellum, and CPG.
- M1_L5B_PT activity drove antagonist extensor-flexor muscles.
- Body motion was fed back to S1, closing the loop.

RESULTS

- M1 output generated reciprocal muscle activation and periodic forearm/lever motion.
- S1 activity was modulated by body motion.
- The loop ran faster than real time overall: 10.05 s in 9.38 s.

CONCLUSION

- GPU execution enabled an overall faster-than-real-time closed-loop brain-body simulation.
- The framework supports testing sensory feedback, cerebellar interactions, and cortico-basal ganglia loops in movement generation.

REFERENCES

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